

● GDG On Campus ● ● ● | Open Innovation Challenge

SMART ATTENDANCE & CLASSROOM UTILIZATION SYSTEM

AI-Powered Face Recognition & Real-Time Analytics

Team: Ctrl+Alt+Defeat

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GLA University, Mathura



The Challenge :

- Manual attendance is time-consuming & error-prone (35-45% proxy rate, 7.5 mins/class)
- No real-time visibility into classroom utilization
- (833+ hours lost/semester)
- Lack of data-driven insights for resource optimization
- Inefficient room allocation & scheduling
- 5-10 mins wasted per class on manual roll call



Our Solution:

AI-Powered Face Recognition Attendance

Real-time biometric verification with instant classroom utilization analytics, eliminating proxy attendance and providing data-driven infrastructure optimization.



SECURE IDENTITY

Face Recognition (94.8%)



SMART ANALYTICS

Gemini AI insights



INSTANT VERIFICATION

Real-time detection



How We Are Different From Others

Traditional QR Codes

Can be shared, screenshot
hacking possible

GPS/Geo-Location

Unreliable indoors, location
spoofing possible

Manual Registers

Time-wasting, error-prone, no analytics

Our Face Recognition

Biometric = Cannot be faked or shared

Face + Room Mapping

Verifies actual presence in specific
room (99.2%)

Automated + AI Analytics

30 seconds to 2+ mins insights for
planning

Key Features



Multi-Frame Face Encoding (94.8% Accuracy)

Captures 5 frames, averages encodings for robust recognition in varied lighting



Real-Time Dashboard with WebSocket Sync

Live attendance tracking across multiple devices simultaneously



Gemini AI-Powered Insights & Recommendations

Predictive analytics for room optimization, schedule planning, & infrastructure decisions

Google Technologies Stack

Gemini AI 2.0



Real-time analytics & predictive insights for campus optimization

Firebase Realtime DB



Attendance records & student data storage

Firebase Cloud Storage



Biometric face database & backups

OpenCV + Face API



Face detection, recognition & verification



Future Enhancements

Google ML Kit



On-device face detection & liveness verification

Firebase Authentication



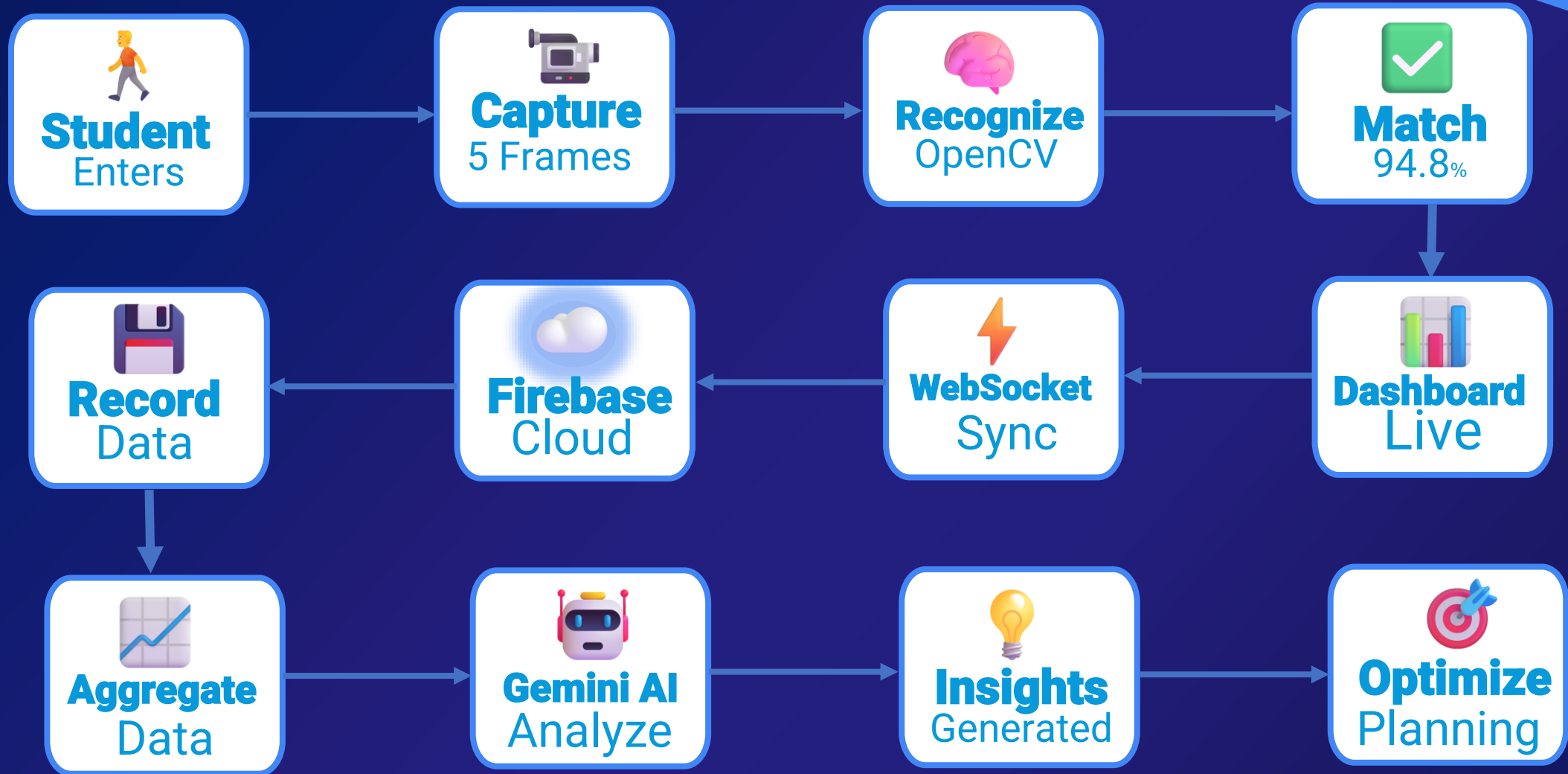
OAuth & Google Sign-In for secure faculty access

Vertex AI



Predictive analytics & smart timetable recommendations

Process Flow Diagram



System Architecture



Teacher Portal



Student Portal



Mobile App

Backend (Flask | OpenCV | Gemini AI)

OpenCV
Detection

Recognition
94.8%

Gemini AI
Analytics

Data Layer (Firebase)

Realtime DB
Records

Cloud Storage
Face DB

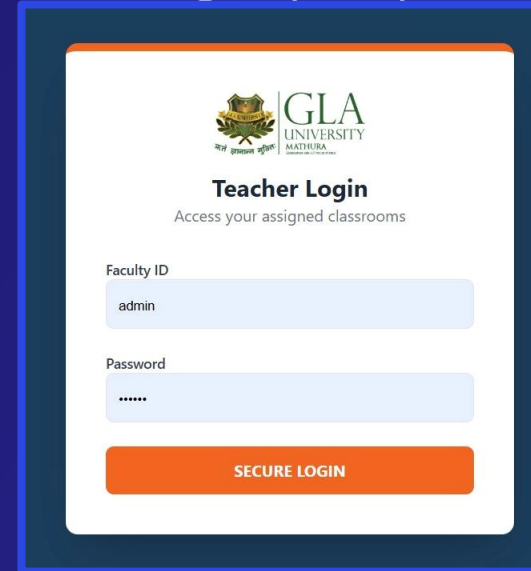
WebSocket
Sync

Login Wireframe

Teacher Login Screen

- GLA University Header
- Faculty ID Input
- Password Input
- "Secure Login" Button

Teacher Login (Live)



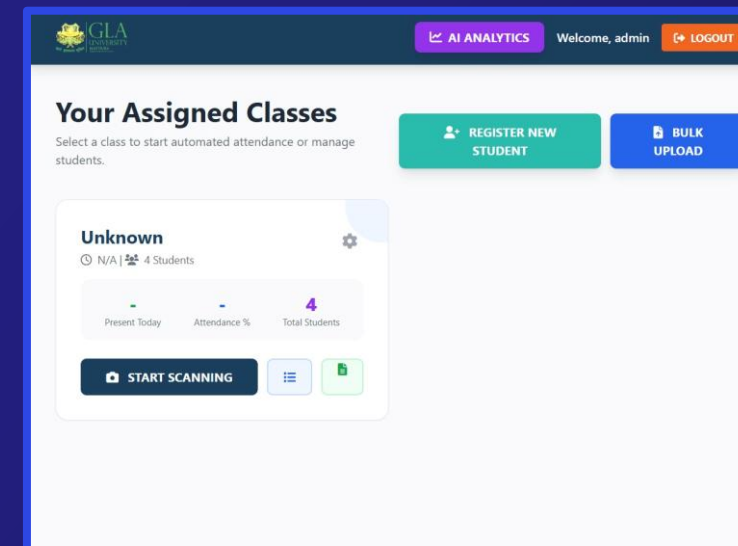
The live Teacher Login screen features the GLA University logo at the top. Below the logo, the text "Teacher Login" is displayed, followed by the subtitle "Access your assigned classrooms". The form includes two input fields: "Faculty ID" with the value "admin" and "Password" with masked characters "*****". A prominent orange button labeled "SECURE LOGIN" is positioned at the bottom of the form.

Dashboard Wireframe

Main Dashboard

- Your Assigned Classes
- [Class Cards] Buttons:
 - START SCANNING
 - ATTENDANCE LIST
- [REGISTER NEW] [BULK UPLOAD]

Dashboard (Live)



The live Main Dashboard features a top navigation bar with the GLA University logo, a purple "AI ANALYTICS" button, a "Welcome, admin" message, and an orange "LOGOUT" button. The main content area is titled "Your Assigned Classes" with the instruction "Select a class to start automated attendance or manage students." To the right of this title are two buttons: "REGISTER NEW STUDENT" and "BULK UPLOAD". Below the title, a class card for "Unknown" is displayed, showing "N/A" for the class name and "4 Students" for the total count. The card includes a "Present Today" section with a green bar, an "Attendance %" section with a blue bar, and a "Total Students" section with a purple bar and the number "4". At the bottom of the card are three buttons: "START SCANNING", a list icon, and a document icon.



Enrollment Wireframe



Student Enrollment

- Roll Number Input
- Full Name Input
- Select Class
- [CAPTURE & SAVE] Button



Student Enrollment (Live)

The live form titled "Student Enrollment" contains the following elements:

- Input field for "University Roll Number (e.g., 21004005)".
- Input field for "Full Name" with a placeholder text "Please fill in this field."
- Input field for "Roll Number (Optional)".
- Dropdown menu for "Select Class" with "-- Select Class --" as the selected option.
- A camera icon and text indicating "Camera will capture face photo automatically on submit".
- A dark blue button labeled "CAPTURE & SAVE".



Analytics Wireframe

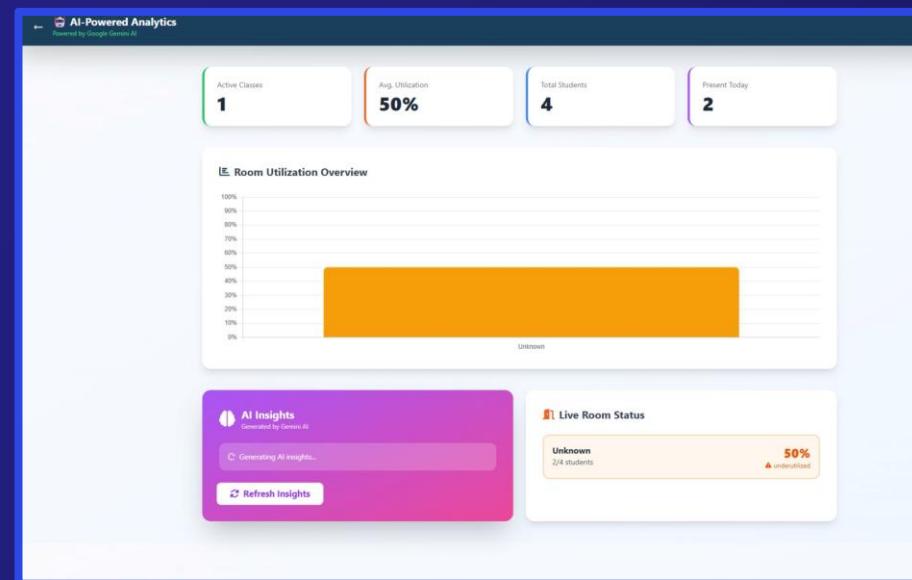


AI Analytics Page

- [KPI Cards]
- Active Classes
- Utilization %
- Room Utilization Chart
- AI Insights (Gemini)



AI Analytics (Live)



Impact & Future

90%

Time Saved

500+

Students/Day

94.8%

Accuracy



Phase 2: Expansion

Scale to 1000+ students • Advanced ML models • Predictive scheduling • ERP integration





THANK YOU
